Round 41: The Octic Order-≥ 24 Census and the R3/R4 Decisive Strikes

GST Hodge Campaign — research-only

2026-11-07 (session clock)

1 The structure theorem (the census engine)

Let (G, ρ) be an octic CM world: G transitive on the 8 embedding letters, ρ a central fixed-point-free involution (complex conjugation). Every such world is S_8-conjugate to one with $\rho = \rho_0 = (04)(15)(26)(37)$, hence $G \subseteq W := C_{S_8}(\rho_0) \cong (C_2)^4 \rtimes S_4$, the hyperoctahedral group of order 384: letters are $(i, e) \in \{0, \ldots, 3\} \times \{0, 1\}$ and $(f, s) \in \mathbb{F}_2^4 \times S_4$ acts by

$$(i, e) \mapsto (s(i),\ e \oplus f_i).$$

Writing $\pi : W \to S_4$ for the pair map,

$$T := \pi(G) \text{ (transitive quartic)}, \qquad K := G \cap (C_2)^4 \ (T\text{-stable},\ \rho_0 \in K),$$

the quotient G/K is a complement of $N' := (C_2)^4/K$ in W_T/K, and conversely every $(T, K, \text{complement})$ yields a world. Since every transitive quartic T is 2-generated, so is every complement; the generating-pair lift sweep is therefore **exhaustive**, and $|G| = |K| \cdot |T|$.

Theorem 1 (The census)**.** *The octic (G, ρ_0)-worlds number* 114*, of which* 58 ***have order*** ≥ 24 *(the Round-37 boundary), reducing to* 24 *W-conjugacy classes, with orders exactly* $\{24, 32, 48, 64, 96, 128, 192, 384\}$.

2 The laws at the boundary

Theorem 2 (Two-algorithm certification)**.** *At every boundary class representative, every type orbit, every degree $2p \leq 8$: the orbit-count law $\dim \mathrm{Hdg}^{2p} = \#P_\Phi(p)$ agrees with the exact blockwise RREF of the equivariance system (*145/145 *bounded checks).*

Theorem 3 (The descent calibrations extend)**.** *At the boundary worlds: the split law* $\dim = \binom{4}{p}^2$ *at every quadratic-inducible type (the* ***exhaustive block classifier****: subgroups $H' \supseteq H$ correspond to blocks containing the letter; the* 128*-subset sweep is complete); the product-world rows* $[8, 18, 8, 1]$ *at the deeper-induced types; the Rosati law $NS = g$ at every primitive type; the palindrome and top laws; the $E \times A$ disjoint-frame tensor law* $\dim \mathrm{Hdg}_2(E \times A) = 1 + NS(A)$*, computed at the genuine joint* 10*-letter worlds.*

Theorem 4 (THE EXCEPTIONAL-CLASS LAW — the round's discovery)**.** *At the intermediate-orbit primitive types (orbit sizes* 4–6*) of the order-*24/48 *worlds, the dimension row is* $[4, 8, 4, 1]$ *and the with-replacement divisor-product ring has rank* 6 *in degree* 4*:*

$\text{surplus} = [0, 2, 0, 0]$ *exactly :* ***two Hodge classes in degree*** 4 ***beyond the divisor ring.***

The Round-37 pairing-generation theorem does not extend past the census boundary; the boundary carries the CM-side analogue of the Weil one-source leaf (R38–R40): the algebraicity of the two exceptional classes is the new falsifiable line.

3 The R3 strike: the general input

Theorem 5 (The vanishing)**.** *The mixed (p,p)-supply on $E \times X$ at the general input — the $(MT(E) \times GSp(M))$-invariant homomorphisms $T : H^1(E) \to M$ — is zero: computed exactly (Fraction nullspace of the stacked equivariance system) at $h = 1, 2$ for both E-models. The mixed sector at the general input is $\{0\}$; the K-existence follows from the Round-32 Theorem-2 equivalence's proven $(a) \Rightarrow (b)$ direction.*

Theorem 6 (The stratum sensitivity)**.** *At the CM-matched stratum ($E[i]\times$ the $\mathbb{Q}(i)^h$-world) the intertwiners number $2h$ exactly (the graph supply, algebraic); at the mismatched strata ($E[i]$ vs $\mathbb{Q}(\sqrt{-d})$, $d = 2, 3, 5$): zero.*

4 The R4 strike: the phase gate

Theorem 7 (The Weil-operator gate)**.** *The two-level middle M_τ in the standard polarization has Weil matrix*

$$C(\tau) = \begin{pmatrix} is/D & -2i/D \\ 2iq/D & -is/D \end{pmatrix}, \qquad D^2 = s^2 - 4q, \quad s = \tau + \bar{\tau}, \quad q = \tau\bar{\tau},$$

*with $C^2 = -I$. The entries are rational exactly when $4q - s^2$ is a rational square (the Gaussian worlds); the CM worlds carry rational Weil operators in the adapted structures ($\left(\begin{smallmatrix} 0 & -d \\ 1 & 0 \end{smallmatrix}\right) \in M_2(\mathbb{Q})$); the entries are nonconstant rational functions of (s, q) — **the transcendental inputs carry no rational Weil operator**.*

Theorem 8 (The phase-order dichotomy and the pinning)**.** *The fold-birth applicability locus is the finite-order-phase world (the exact orders $2, 3, 4, 6$ — the Round-40 collar, including $\tau = 1+i$ (order 4) and $\tau = 3+\sqrt{-3}$ (order 6)); the phase $7+4\sqrt{3}$ has infinite order ($z^n \neq 1$, exact $\mathbb{Z}[\sqrt{3}]$ arithmetic); $t^n - 1 \neq 0$ as a polynomial: the root-of-unity locus is a proper subvariety. Hence:*

R4's irreducible content = the transcendental (p,p)-algebraicity,

*and R3's residue is the same face: **the input axis reduces to ONE irreducible core** (the Round-32 Theorem-4 equivalence's two named constructions now one).*

5 The certificates

1. `octic_census_round41_checks.py`: 50 checks, 0 failures, exit 0, deterministic double-run byte-identical (md5 $d0b54dac4b95b6829b2ed6060edf3e82$); exact arithmetic, zero floats; the disclosed bounds: the product battery at $|G| \leq 96$, the RREF systems at ≤ 3000 variables, the cocycle sweep generator-based above $|G| = 96$.

2. `r3_r4_strike_round41_checks.py`: 33 checks, 0 failures, exit 0, byte-identical double-run (md5 $40128c51996cbbb28518dc9e4334ad1c$).

The AUDIT-41 loop: PASS_WITH_REPAIRS_REQUIRED → the F1 repair (the deeper-bucket law strengthened: the binomial row cannot be deeper-induced) → re-sealed green → the previously-surviving sabotage re-fired → CLOSED at PASS.

The quantifier line

The octic order-≥ 24 CM-transport boundary is enumerated exhaustively and the descent laws certified at every world; the exceptional-class law measured exactly; the input axis's two named constructions struck decisively and merged into one irreducible core. The remaining falsifiable line: the transcendental face + the two exceptional-class leaves + the CM transport beyond degree 8. **Q2 not claimed; the closing string not emitted; the polarity split stands.**